KECU Interface Message Protocol Specification

Document History

Date	Version	Author	Description
2021-05-17	V0.1	Ismo Kaunisto	Initial version
2021-06-01	V0.2	Ismo Kaunisto	Typo fixes, added notifications NTF_NODE_INSERTED and NTF_NODE_REMOVED
2022-01-10	V0.3	Ismo Kaunisto	HV source commands and notification added
2022-01-25	V0.4	Ismo Kaunisto	Valve commands and notification added

1 Introduction

This document describes the message protocol used in Karsa KECU project. Messages are delivered between application (APP) and KECU FW (FW) using Ethernet interface with TCP and UDP sockets. TCP server running in FW uses by default fixed IP address 192.168.1.200. Command and reply messages are using TCP port 65142. High priority notification messages are using TCP port 65143 and measurement data notifications are using by default UDP port 65146.

1.1 Definitions, Acronyms and Abbreviations

AI	Analog Input device (uCAN.6.ai-SNAP)
APP	Measurement and Control application in PC
CAN	Controller Area Network
DIO	Digital Input and Output device (uCAN.8.dio.SNAP)
ETX	End of Text
FW	Firmware in KECU hardware unit
HV	High voltage
MFC	Mass Flow Controller (Bürkert 8741/8742)
STX	Start of Text

2 Message formats

All messages are encapsulated into a transfer frame. The transfer frame consists of STX and ETX markers. Length of STX and ETX is 8 bits. The STX marker is added at the beginning of the message and ETX at the end of the message.

STX	MESSAGE	ETX

The frame marker defines:

STX = 0x02h

ETX = 0x03h

Generic message header is shown below. Type field identifies the message. The number of bytes received after length field is length + 1 (ETX). All fields in the header are 8-bits.

STX	TYPE	LENGTH

Reply messages contains status field. All values, other than zero, are error values. The Type field in the reply message is same than in command message.

2.1 Command message format

STX	TYPE	LENGTH	PAYLOAD (0 – N bytes)	ETX

LENGTH : Payload length

TYPE: bit 7: 0 – command from APP to FW

2.2 Reply message format

STX	TYPE	LENGTH	STATUS	PAYLOAD (0 – N bytes)	ETX

LENGTH : Payload length + 1

TYPE: bit 7: 0 – reply from FW to APP

2.3 Notification message format

STX	TYPE	LENGTH	PAYLOAD (0 – N bytes)	ETX

LENGTH : Payload length

TYPE: bit 7: 1 – Measurement data from FW to APP

No reply is sent to the notification message.

3 Messages

3.1 Notification messages

3.1.1 NTF_AI_MEAS_DATA_CH_1_4

Measurement data from AI unit, channels 1-4.

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x80
LENGTH	1 byte	Length of the payload	0x09
Node Id	1 byte	Device node identifier	
AI Process Value of Channel 1	2 bytes, LSB first	Signed 16-bit value	
AI Process Value of Channel 2	2 bytes, LSB first	Signed 16-bit value	

AI Process Value of Channel 3	2 bytes, LSB first	Signed 16-bit value	
AI Process Value of Channel 4	2 bytes, LSB first	Signed 16-bit value	
ETX	1 byte	marker	0x03

3.1.2 NTF_AI_MEAS_DATA_CH_5_6

Measurement data from AI unit, channels 5-6.

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x81
LENGTH	1 byte	Length of the payload	0x05
Node Id	1 byte	Device node identifier	
AI Process Value of Channel 5	2 bytes, LSB first	Signed 16-bit value	
AI Process Value of Channel 6	2 bytes, LSB first	Signed 16-bit value	
ETX	1 byte	marker	0x03

3.1.3 NTF_DIO_MEAS_DATA

Measurement data from DIO unit

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x82
LENGTH	1 byte	Length of the payload	0x02
Node Id	1 byte	Device node identifier	
Digital Input Value	1 byte		
ETX	1 byte	marker	0x03

3.1.4 NTF_MFC_MEAS_DATA

Measurement data from MFC unit

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x83
LENGTH	1 byte	Length of the payload	0x05 – 0x08
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
Data value	1 – 4 bytes, LSB first	Data value size depends on Object and Sub values (= LENGTH – 4)	
ETX	1 byte	marker	0x03

3.1.5 NTF_HV_MEAS_DATA

Measurement data from HV source unit

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x84
LENGTH	1 byte	Length of the payload	0x06
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
channel	1 byte	Channel number	0x01 – 0x03
Data value	2 bytes, LSB first	Signed 16-bit value	
ETX	1 byte	marker	0x03

3.1.6 NTF_VALVE_MEAS_DATA

Measurement data from Valve unit

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x85
LENGTH	1 byte	Length of the payload	0x05 – 0x08
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
Data value	1 – 4 bytes, LSB first	Data value size depends on Object and Sub values (= LENGTH – 4)	
ETX	1 byte	marker	0x03

3.1.7 NTF_NODE_INSERTED

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x90
LENGTH	1 byte	Length of the payload	0x02
Node Id	1 byte	Device node identifier	
Node Type	1 byte	Type of the node	
ETX	1 byte	marker	0x03

3.1.8 NTF_NODE_REMOVED

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x91
LENGTH	1 byte	Length of the payload	0x01
Node Id	1 byte	Device node identifier	
ETX	1 byte	marker	0x03

3.2 COMMAND MESSAGES

3.2.1 CMD_READ_FW_VERSION

This message is used to get FW version.

CMD_READ_FW_VERSION request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x01
LENGTH	1 byte	Length of the payload	0x00
ETX	1 byte	marker	0x03

CMD_READ_FW_VERSION reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x01
LENGTH	1 byte	Length of the payload	0x03
STATUS	1 byte	0x00 - OK	0x00
Major Version	1 byte		
Minor Version	1 byte		
ETX	1 byte	marker	0x03

3.2.2 CMD_RESET_NODE

CMD_RESET_NODE request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x02
LENGTH	1 byte	Length of the payload	0x01
Node Id	1 byte	Device node identifier	
ETX	1 byte	marker	0x03

Note: If Node Id is 0x00, reset command is sent to all nodes found in CAN network.

CMD_RESET_NODE reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x02
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.3 CMD_GET_NODE_LIST

CMD_GET_NODE_LIST request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x03
LENGTH	1 byte	Length of the payload	0x00
ETX	1 byte	marker	0x03

CMD_GET_NODE_LIST reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x03
LENGTH	1 byte	Length of the payload 2 * Nodes found + 1	
STATUS	1 byte	0x00 - OK	0x00
Node Id	1 byte	Device node identifier 1	
Node Type	1 byte	Type of the node 1	
Node Id	1 byte	Device node identifier 2	
Node Type	1 byte	Type of the node 2	
. . .			
Node Id	1 byte	Device node identifier n	
Node Type	1 byte	Type of the node n	
ETX	1 byte	marker	0x03

3.2.4 CMD_GET_NODE_DATA

CMD_GET_NODE_DATA request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x04
LENGTH	1 byte	Length of the payload	0x04
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
ETX	1 byte	marker	0x03

CMD_GET_NODE_DATA reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x04

LENGTH	1 byte	Length of the payload	0x02 – 0x21 (OK) 0x01 (STATUS NOT OK)
STATUS	1 byte		
Data	N bytes	Max 32 bytes	
ETX	1 byte	marker	0x03

3.2.5 CMD_SET_NODE_DATA

CMD_SET_NODE_DATA request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x05
LENGTH	1 byte	Length of the payload	0x05 – 0x08
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
Data	1 – 4 bytes		
ETX	1 byte	marker	0x03

CMD_SET_NODE_DATA reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x05
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.6 CMD_START_MFC_MEAS

CMD_START_MFC_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x10
LENGTH	1 byte	Length of the payload	0x05
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
Interval	1 byte	Bit 0 – 6: Measurement interval in 100ms Bit 7: Measurement data notification port selection.0 – TCP,1 - UDP	
ETX	1 byte	marker	0x03

Note: If Node Id is 0x00, measurement is started for all MFC nodes found in CAN network.

CMD_START_MFC_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x10
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.7 CMD_STOP_MFC_MEAS

CMD_STOP_MFC_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x11
LENGTH	1 byte	Length of the payload	0x04
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
ETX	1 byte	marker	0x03

Note1: If Node Id is 0x00, measurement is stopped for all MFC nodes found in CAN network.

Note2: If Index is 0x0000 and sub-index is 0x00, all measurements are stopped from the MFC node.

CMD_STOP_MFC_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x11
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.8 CMD_START_AI_MEAS

CMD_START_AI_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x12
LENGTH	1 byte	Length of the payload	0x03
Node Id	1 byte	Device node identifier	
Channel mask	1 byte	Bit 0: 1 – channels 1 - 4 Bit 1: 1 – channels 5 - 6	

Interval	1 byte	Bit 0 – 6: Measurement interval in 100ms Bit 7: Measurement data notification port selection.0 – TCP,1 - UDP	
ETX	1 byte	marker	0x03

CMD_START_AI_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x12
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.9 CMD_STOP_AI_MEAS

CMD_STOP_AI_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x13
LENGTH	1 byte	Length of the payload	0x02
Node Id	1 byte	Device node identifier	
Channel mask	1 byte	Bit 0: 1 – channels 1 - 4 Bit 1: 1 – channels 5 - 6	
ETX	1 byte	marker	0x03

CMD_STOP_AI_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x13
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.10 CMD_START_DIO_MEAS

CMD_START_DIO_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x14
LENGTH	1 byte	Length of the payload	0x02
Node Id	1 byte	Device node identifier	

Interval	1 byte	Bit 0 – 6: Measurement interval in 100ms Bit 7: Measurement data notification port selection.0 – TCP,1 - UDP	
ETX	1 byte	marker	0x03

CMD_START_DIO_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x14
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.11 CMD_STOP_DIO_MEAS

CMD_STOP_DIO_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x15
LENGTH	1 byte	Length of the payload	0x01
Node Id	1 byte	Device node identifier	
ETX	1 byte	marker	0x03

CMD_STOP_DIO_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x15
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.12 CMD_START_HV_MEAS

CMD_START_HV_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x16
LENGTH	1 byte	Length of the payload	0x05
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	

		(0x7130, 0x7131 or 0x7300)	
Channel mask	1 byte	Bit 0: 1 – channel 1 Bit 1: 1 – channel 2 Bit 2: 1 – channel 3	0x01 – 0x07
Interval	1 byte	Bit 0 – 6: Measurement interval in 100ms Bit 7: Measurement data notification port selection.0 – TCP,1 - UDP	
ETX	1 byte	marker	0x03

Note: If Node Id is 0x00, measurement is started from the channels specified in the channel mask of all HV source nodes found in CAN network.

CMD_START_HV_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x16
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.13 CMD_STOP_HV_MEAS

CMD_STOP_HV_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x17
LENGTH	1 byte	Length of the payload	0x04
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object (0x7130, 0x7131,0x7300 or 0x0000)	
Sub-index	1 byte		
ETX	1 byte	marker	0x03

Note1: If Node Id is 0x00, measurement is stopped from the channels specified in the channel mask of all HV source nodes found in CAN network.

Note2: If Index is 0x0000, all measurements are stopped from the HV source node channels specified in the channel mask.

CMD_STOP_HV_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)

STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x17
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.14 CMD_START_VALVE_MEAS

CMD_START_VALVE_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x18
LENGTH	1 byte	Length of the payload	0x05
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
Interval	1 byte	Bit 0 – 6: Measurement interval in 100ms Bit 7: Measurement data notification port selection.0 – TCP,1 - UDP	
ETX	1 byte	marker	0x03

Note: If Node Id is 0x00, measurement is started for all Valve nodes found in CAN network.

CMD_START_VALVE_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x18
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

3.2.15 CMD_STOP_VALVE_MEAS

CMD_STOP_VALVE_MEAS request message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x19
LENGTH	1 byte	Length of the payload	0x04
Node Id	1 byte	Device node identifier	
Index	2 bytes, LSB first	Address of the object	
Sub-index	1 byte		
ETX	1 byte	marker	0x03

Note1: If Node Id is 0x00, measurement is stopped for all Valve nodes found in CAN network.

Note2: If Index is 0x0000 and sub-index is 0x00, all measurements are stopped from the Valve node.

CMD_STOP_VALVE_MEAS reply message:

Field	**Size**	**Comment**	**Value** (if known)
STX	1 byte	marker	0x02
TYPE	1 byte	Type of the message	0x19
LENGTH	1 byte	Length of the payload	0x01
STATUS	1 byte		
ETX	1 byte	marker	0x03

4 Error codes

KRS_SUCCESS	0x00
KRS_ERR_UNKOWN_COMMAND	0x01
KRS_ERR_INVALID_PARAM	0x02
KRS_ERR_INVALID_LENGTH	0x03
KRS_ERR_INTERNAL	0x04
KRS_ERR_NODE_NOT_SUPPORTED	0x10
KRS_ERR_NODE_NOT_FOUND	0x11
KRS_ERR_INVALID_NODE_TYPE	0x12
KRS_ERR_CAN_TX	0x20
KRS_ERR_CAN_RX	0x21
KRS_ERR_UNSUPPORTED_DATASIZE	0x22
KRS_ERR_MEAS_NOT_ONGOING	0x30
KRS_ERR_INVALID_OBJ_SUB_PAIR	0x31
KRS_ERR_NO_MEAS_ENTRY	0x32
KRS_ERR_MEAS_START	0x40
KRS_ERR_AI_MEAS_START_CH_1_4	0x41
KRS_ERR_AI_MEAS_START_CH_5_6	0x42
KRS_ERR_AI_MEAS_START_CH_1_6	0x43

5 Defines

5.1 Node types

0	Bürkert 8741/8742 MFC
1	Digital Input and Output Module
2	Analog Input Module
3	Bürkert 0331 Valve
4	Karsa high voltage source

255	Unknown type